

Quantum Computation and Information

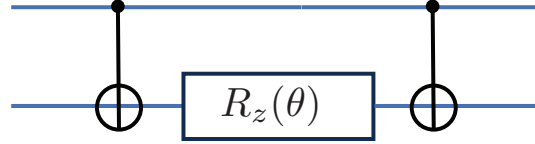
Homework Set 2

1. Recall Pauli X, Y, Z matrices: $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

- (a). Show that for any $j \in \{1, 2, 3\}$

$$e^{-i\theta\sigma_j} = I \cos \theta - i\sigma_j \sin \theta$$

- (b). Consider a system of two qubits. Denote by Z_0 and Z_1 the Pauli Z matrices on qubits 0 and 1, respectively. Show that the unitary transformation $U(\theta) = e^{-i\theta Z_0 Z_1}$ can be implemented by the following circuit with $R_z(\theta) = e^{-i\theta\sigma_3}$.



2. **(Amplitude Amplification)** Given a unitary operator U on a system of n qubits:

$$U |0^{\otimes n}\rangle = \cos \theta |\Psi_0\rangle + \sin \theta |\Psi_1\rangle,$$

where $\langle \Psi_0 | \Psi_0 \rangle = \langle \Psi_1 | \Psi_1 \rangle = 1$ and $\langle \Psi_0 | \Psi_1 \rangle = 0$. Define $S_1 = I - 2 |\Psi_1\rangle \langle \Psi_1|$, $S_i = I - 2 |0^{\otimes n}\rangle \langle 0^{\otimes n}|$, and $Q = -US_iU^{-1}S_1$.

- (a). Show that

$$\begin{aligned} Q |\Psi_1\rangle &= \cos 2\theta |\Psi_1\rangle - \sin 2\theta |\Psi_0\rangle \\ Q |\Psi_0\rangle &= \sin 2\theta |\Psi_1\rangle + \cos 2\theta |\Psi_0\rangle \end{aligned}$$

The above relations implies the two-dimensional space spanned by $|\Psi_0\rangle$ and $|\Psi_1\rangle$ is invariant under operator Q .

- (b). Show that

$$Q^k U |0^{\otimes n}\rangle = \cos (2k + 1)\theta |\Psi_0\rangle + \sin (2k + 1)\theta |\Psi_1\rangle, \quad k \in \mathbb{Z}_+.$$

Suppose θ is very small. If $|\Psi_1\rangle$ is the desired result, the above relation implies that the probability of obtaining the desired result increases quadratically with the number of times Q is applied.

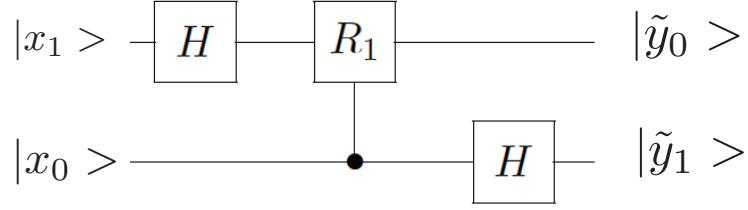
3. Consider the quantum Fourier transformation

$$QFT_{2^2} |x\rangle = \frac{1}{\sqrt{2^2}} \sum_{y=0}^{2^2-1} e^{i\frac{2\pi xy}{2^2}} |y\rangle = \frac{1}{\sqrt{2^2}} (|0\rangle + e^{i2\pi(.x_0)} |1\rangle)(|0\rangle + e^{i2\pi(.x_1x_0)} |1\rangle).$$

(a). Recall the notation introduced in lecture 3: $x \equiv x_1x_0$ and $y \equiv y_1y_0$. Show that

$$\frac{xy}{2^2} \pmod{1} \equiv y_1(.x_0) + y_0(.x_1x_0)$$

(b). Show that QFT_{2^2} can be computed by the following circuit:



where $|\tilde{y}_0\rangle = |0\rangle + e^{i2\pi(.x_1x_0)} |1\rangle$, $|\tilde{y}_1\rangle = |0\rangle + e^{i2\pi(.x_0)}$, and $R_1 = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{2}} \end{pmatrix}$.